

Pranavesh Kumar Talupuri

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Education

- University of Southern California**, Master of Science in Computer Science Aug 2025 – May 2027
- GPA: 3.75 (Tentative)
 - Coursework: Deep Learning, 3D Graphics and Rendering, Analysis of Algorithms, Database Systems
- Vellore Institute of Technology - AP**, B.Tech. in Computer Science and Engineering Aug 2021 – May 2025
(Specialization in Artificial Intelligence and Machine Learning)
- GPA: 9.0/10
 - Relevant coursework: Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Reinforcement Learning, Digital Image Processing

Research Experience

- Graduate Research Assistant**, University of Southern California May 2026 – Current
- Machine Learning and Differential Privacy research with Prof. Sai Praneeth Karimireddy, working closely with Mohammadamin Banayeeanzade and Tony Yang
- Graduate Research Assistant**, University of Southern California Feb 2026 – Apr 2026
- Developed a deep learning methodology for identifying melanoma-afflicted cells from biomedical imagery.
- Machine Learning Researcher (Intern)**, Virginia Tech Research Forum Jan 2025 – Jun 2025
- Developed a machine learning model for accelerated discovery of high performance piezoelectric materials.
 - Built computer vision and deep learning models for an autonomous underwater vehicle, contributing to perception for robotics applications.
- Undergraduate Research Assistant / Project Lead**, Vellore Institute of Technology Oct 2022 – May 2025
- Led a team of 7 student engineers to design, build, and test a functional VTOL prototype.
 - Initiated computer vision applications on the VTOL platform, including surveillance and survey use cases.
 - Worked across hardware prototyping and perception software for robotics systems and applied vision research.

Publications

- Analysis of Acute Lymphoblastic Leukemia Detection Methods Using Deep Learning**
Pranavesh Kumar Talupuri, Beebi Naseeba, Nagendra Panini Challa, Abbaraju Sai Sathwik.
2023 International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS), 2023.
doi:10.1109/ICSSAS57918.2023.10331776
Reviews deep learning approaches for acute lymphoblastic leukemia detection for image based diagnosis.
- Smart MCQ Generation using KeyBERT and BERT Based Models**
Pranavesh Kumar Talupuri, Manjunaatt Girish Kumar, Kotha Lavanya, Gudditi Chetan, Nagendra Panini Challa.
2024 3rd Edition of IEEE Delhi Section Flagship Conference (DELCON), 2024. doi:10.1109/DELCON64804.2024.10866615
System for automatic multiple choice question generation using keyword extraction and language modeling using BERT.
- Hybrid Time Series Forecasting with ARIMA, Echo State Networks, and LSTM**
Sai Eeshwar D, Pranavesh Kumar Talupuri, Aarush Shivkumar, Beebi Naseeba.
2025 International Conference on Computing Technologies & Data Communication (ICCTDC), 2025.
doi:10.1109/ICCTDC64446.2025.11158940
Proposes a hybrid forecasting model that combines ARIMA, echo state networks, and LSTMs with adaptive weighting to improve predictive accuracy.
- Comparison of Deep Learning Models for Classification of Acute Lymphoblastic Leukemia**
Pranavesh Kumar Talupuri, Nitin Bhimineni, Beebi Naseeba, E. R. Aruna, Sai Vennela Tipirneni, Nagendra Panini Challa.
Pertanika Proceedings, 1(5):93–99, 2025. doi:10.47836/pp.1.5.018
Compares CNN architectures for classifying acute lymphoblastic leukemia from cell images and studying tradeoffs across model families.

Selected Research Projects

Efficient 3D Gaussian Splatting for Real-Time Rendering and Reconstruction using Adaptive Level of Detail

- Developed a real-time 3D Gaussian splatting pipeline for rendering and scene reconstruction, with an emphasis on efficiency and scalable scene representation.
- Implemented adaptive LOD strategies in OpenGL/GLSL to reduce rendering cost while preserving visual fidelity across viewpoints.
- Achieved 50%–90% frame-rate improvement on benchmark scenes through dynamic pruning and per-frame rendering optimization.

Hybrid 3D/4D Gaussian Splatting for Scene Reconstruction

- Created an adaptive hybrid 3D/4D Gaussian splatting framework for blur aware reconstruction of large-scale dynamic outdoor scenes (Waymo Open).
- Achieved real time framerates matching those of benchmark.

Adaptive Matryoshka Embeddings for Material and Graphics Asset Retrieval

- Built an OpenCLIP retrieval pipeline for material, texture, and graphics asset searching using nested embeddings, learned projection heads, and coarse-to-fine reranking.
- Implemented a benchmarking and analysis stack with metadata retrieval metrics, qualitative retrieval grids, and cross dataset evaluation on texture and material benchmarks.
- Evaluated tradeoffs of memory and latency against CLIP truncation, PCA, and learned projection baselines, showing that 32-dimensional Matryoshka embeddings preserved strong retrieval quality while reducing embedding memory by 93.8% (16× smaller) relative to 512-dimensional full CLIP on FMD.

Multi Decadal Land Use and Land Cover Change Analysis using Google Earth Engine

- Developed a cloud based remote sensing pipeline in Google Earth Engine to classify land use and land cover across three decades: 1985, 1995, and 2005.
- Multi sensor satellite imagery from Landsat, Sentinel, and MODIS was used to analyze long term spatial changes in vegetation, water bodies, built up land, agricultural regions, barren land, and forest classes.
- Spectral and vegetation features including NDVI, NDWI, NDBI, EVI, SAVI, MSAVI, and spectral transform bands were engineered to improve class separation across heterogeneous land cover categories.
- Supervised classifiers including Random Forest, SVM, CART, and Gradient Tree Boosting were trained and compared, with model outputs combined through an ensemble based classification workflow.
- Temporal change analysis was performed using stratified sampling, class consolidation, spatial post processing, accuracy assessment, kappa evaluation, area statistics, and exportable LULC maps.

Hybrid YOLO and Hough Transform Framework for Rural Road Edge Detection

- Developed a vision based road edge detection framework for unstructured rural roads without painted lane markings, targeting use cases in low infrastructure driving environments.
- Combined YOLO scene object detection with classical Hough transform geometry to identify road boundaries from visual cues such as edges, shoulders, vegetation transitions, and terrain structure.
- Designed the pipeline to handle challenging outdoor conditions including irregular road shapes, weak boundary contrast, and cluttered rural backgrounds.

Technical Skills

Programming Languages: Python, C, C++, C#, Java, R, SQL, MATLAB

Deep Learning: PyTorch, TensorFlow, OpenCV, scikit-learn, Hugging Face, Ultralytics

Tools: Linux, Git, CUDA, LaTeX

Exam Scores

GRE: 325/340 (Quantitative Reasoning : 168/170, Verbal Reasoning : 157/170, Analytical Writing : 4.5)

IELTS: 8.5/9 (Reading : 9, Listening : 9, Speaking : 8, Writing : 7.5)